

CHANDANA

Data Engineer

+1 267-719-3550

iamchandanaach19@gmail.com

PROFESSIONAL SUMMARY:

- Having around **5+** years of experience in the IT sector, the highly motivated, motivating, and skilled data engineer has explored a wide range of technologies, tools, and databases, including **Big Data, AWS, S3, Snowflake, Hadoop, Hive, Spark, Python, Sqoop, Tableau, SQL, PLSQL, and Redshift**. Worked and learned a great deal from Amazon Web Services (AWS) cloud services like EC2, S3, EBS.
- Experience in writing code in Python to manipulate the data in performing statistical analysis, Data load, Data extract & Data Transform etc.
- Designed and implemented robust data pipelines using frameworks like Apache Spark and AWS Glue to process and transform large-scale datasets.
- Strong technical and architectural knowledge in solution design and development, Designed scalable data models and schemas for efficient querying and reporting.
- Used Hive SQL, Presto SQL, and Spark SQL for ETL jobs and using the right technology for the job to get done.
- Automated CI/CD pipelines for deploying data processing jobs, ensuring faster delivery and reduced downtime.
- Implemented data governance frameworks ensuring data accuracy, security, and compliance using AWS Lake Formation with policies enforced via SQL, IAM roles, Python, and Airflow.
- Integrated version control systems like **Git** and **Bitbucket** with CI/CD workflows to enable efficient code reviews and branching strategies.
- Automated infrastructure provisioning and configuration management using **CloudFormation**
- Good Knowledge in all phases of Software Development Life Cycle, Agile and Waterfall methodologies.
- Hands on experience in developing SPARK applications using Spark tools like RDD Transformations, Spark core, Spark Streaming and Spark SQL.
- Experienced in working with in-memory processing frameworks like Spark Transformations, Spark SQL and Spark Streaming.
- Automated workflows using AWS Lambda and Step Functions to improve operational efficiency.
- Optimized cost by implementing right-sizing strategies, Reserved Instances, and leveraging AWS Cost Explorer for monitoring.

TECHNICAL SKILLS:

Big Data	Hadoop, Hive, Oozie, Flume, Spark, PySpark, Spark SQL, HDFS, Yarn, Apache Airflow, Apache Iceberg
Programming Languages	SQL, PL/SQL, C, Python, HiveQL
IDE	Visual Studio code, Notepad++, Spring tool suite, GitHub, Eclipse, Bitbucket, Jupyter
OS	Windows, Linux (CentOS)
Databases	MySQL, Oracle, SQL Server, PostgreSQL
Cloud Tools	AWS Glue, AWS Athena, IAM, S3, EMR, EC2, CloudFormation, Lambda, AWS SageMaker
Methodologies	Agile Software Development, Waterfall

PROFESSIONAL EXPERIENCE:

USAA, Texas
Data Engineer

December 2023-Current

Responsibilities:

- Developed data pipelines for analyzing data.
- Developed Oozie workflow for scheduling and orchestrating the ETL process.
- Designed, developed, and maintained scalable ETL pipelines using AWS Glue for data extraction, transformation, and loading across various data sources.
- Created data lake on Amazon S3. Integrated iceberg tables with cloud storage systems like S3 to manage large datasets effectively.
- Designed and implemented serverless ETL jobs using AWS Glue, optimizing data transformations with Python UDFs and leveraging crawlers for schema discovery.
- Orchestrated complex ETL workflows using AWS Data Pipeline, scheduling jobs, handling dependencies, and monitoring pipeline execution.
- Monitored data quality checks. Designed and executed comprehensive data validation checks to identify and correct errors, inconsistencies, and anomalies within ETL pipelines.
- Designed and built data warehouses on AWS Redshift, optimizing query performance with proper data modeling (star/snowflake schemas), partitioning, and sorting.
- Optimized ETL jobs and queries in Glue and Athena by tuning resource allocation, enabling use of AWS resources and cost management.
- Optimized ETL pipelines to improve performance, reduce processing time, and minimize costs by leveraging AWS services like AWS Glue, AWS EMR.
- Utilized Amazon Athena to perform complex queries on large datasets stored in Amazon S3, optimizing query performance and reducing data processing time.

- Tuned database queries, optimized data storage, and implemented caching mechanisms to improve data retrieval and processing performance.
- Developed Oozie workflow for scheduling and orchestrating the ETL process.
- Configured AWS services including IAM to support PySpark workloads, besides making sure that best practices and organizational security policies are followed.
- Designed and implemented serverless data pipelines using AWS Lambda and Step Functions, enabling auto-scaling and pay-per-use pricing.
- Created and configured CloudWatch alarms to notify on critical events (e.g., high resource utilization, data pipeline failures, data quality issues).
- Optimized ETL jobs and queries in Glue and Athena by tuning resource allocation, enabling use of AWS resources and cost management.
- Created custom dashboards in CloudWatch to visualize key performance indicators and monitor the health of data pipelines.
- Integrated Iceberg tables with cloud storage systems like S3 to manage large datasets effectively.

Environment: PySpark, ETL, S3, Oozie, AWS, IAM, AWS Glue, Athena, Iceberg

American Express, USA
Data Engineer

Jan 2023-Sep 2023

Responsibilities:

- Designed a data warehouse using Hive. Created partitioned tables in Hive. And implemented partitioning and bucketing in Hive.
- Hands on experience in various Big Data application phases like data ingestion, data analytics and data visualization.
- Implemented robust data quality checks and validation rules to ensure data accuracy and consistency, including data profiling, cleansing, and deduplication.
- Tuned database queries and optimized data storage to enhance query performance and minimize resource utilization.
- Developed reusable components and modularized ETL processes to improve code maintainability and reduce development time
- Proficiency in serverless computing platforms, such as AWS Lambda, Azure Functions, for building scalable and cost-effective ETL pipelines.
- Designed and implemented automated responses to critical alerts, such as auto-scaling EC2 instances, triggering re-runs of failed jobs, and notifying on-call teams via appropriate channels.
- Leveraged AWS CloudTrail to audit and monitor API calls made to AWS services, enhancing security posture and identifying potential threats.
- Developed and implemented granular IAM policies using JSON to define fine-grained permissions for AWS resources, minimizing the risk of unintended access.
- Collaborated with security teams to ensure adherence to security best practices and compliance requirements.

Environment: PySpark, ETL, S3, Oozie, AWS (IAM, Glue, Athena, CloudTrail), Hive, Iceberg, Lambda

Responsibilities:

- Developed a data pipeline using Spark and Hive to ingest, transform and analyzing data.
- Build pipelines to work with large datasets using Glue and PySpark. Created, managed, and optimized ETL/ELT processes.
- Utilized Glacier and Glacier Deep Archive for long-term archival storage of infrequently accessed data, significantly reducing storage costs.
- Implemented SSE-S3 or SSE-KMS to encrypt data at rest, ensuring data security and compliance while minimizing performance impact.
- Enabled S3 Storage Analytics to collect and analyze access patterns to S3 objects, providing insights for storage optimization.
- Performed performance tuning and query optimization and assisted with performance tuning, monitoring and troubleshooting.
- Utilized S3 Transfer Acceleration to speed up data transfers to and from S3, improving data ingestion and retrieval performance.
- Leveraged S3 Batch Operations to perform bulk operations on S3 objects, such as copying, moving, and deleting, efficiently and at scale.
- Implemented security measures to protect data integrity and privacy.
- Creating hive tables and working on them for data analysis to meet the business requirements.
- Consistently applied the principle of least privilege to all IAM users, roles, and groups, granting only the necessary permissions to perform their job functions.
- Very well versed in workflow scheduling and monitoring tools such as Oozie.
- Hands on experience working as production support engineer.
- Configured AWS services including IAM to support PySpark workloads, besides making sure that best practices and organizational security policies are followed.
- Designed, developed, and maintained scalable ETL pipelines using AWS glue for data extraction, transformation, and loading across various data sources.
- Utilized Amazon Athena to perform complex queries on large datasets stored in Amazon S3, optimizing query performance and reducing data processing time.
- Implemented data governance policies by managing table-level access control, auditing data changes, and ensuring data integrity with iceberg's ACID compliance.
- Worked on performance tuning to ensure that assigned systems were patched, configured and optimized for maximum functionality and availability.
- Set up FTP process to move files from windows share drive to S3 buckets.
- Suggested and implemented and automated data quality check process which checks for data integrity of data loaded.

Environment: Spark, Hive, PySpark, ETL, SQL, Linux, S3, Oozie, AWS, IAM, Sqoop, AWS Glue, Athena, Iceberg.

Responsibilities:

- Installed and configured database systems (e.g., MySQL, PostgreSQL, Oracle).
- Successfully migrated a critical database to a new server, resulting in a 20% increase in query performance and a 10% reduction in hardware costs.
- Performed regular maintenance tasks such as backups, recovery, and updates.
- Automated database provisioning and configuration processes using tools like Terraform or Ansible, improving deployment speed and reducing human error.
- Monitor database performance and tuned systems for optimal efficiency.
- Implemented security measures to protect databases, including user access controls and encryption.
- Ensure data integrity and consistency through regular checks and audits.
- Developed and managed backup and recovery plans to prevent data loss.
- Diagnosed and resolved database issues and errors.
- Optimized database queries and indexes to enhance performance.
- Created and maintained detailed documentation of database configurations, processes, and procedures.
- Conducted capacity planning to accommodate future database growth.
- Setup and managed database replications and clustering for high availability.
- Ensured compliance with data protection regulations and company policies.
- Automated routine tasks using scripts and database management tools.
- Experience with cloud-based database services like AWS RDS and its administration.
- Successfully migrated on-premises databases to AWS RDS to improve scalability, reduce costs, and enhance performance.

Environment: SQL, MYSQL, PostgreSQL, Oracle, AWS RDS

EDUCATION:

- Masters in computer and information sciences, University of North Texas, USA
- Bachelors in computer science and engineering, SREC, India